

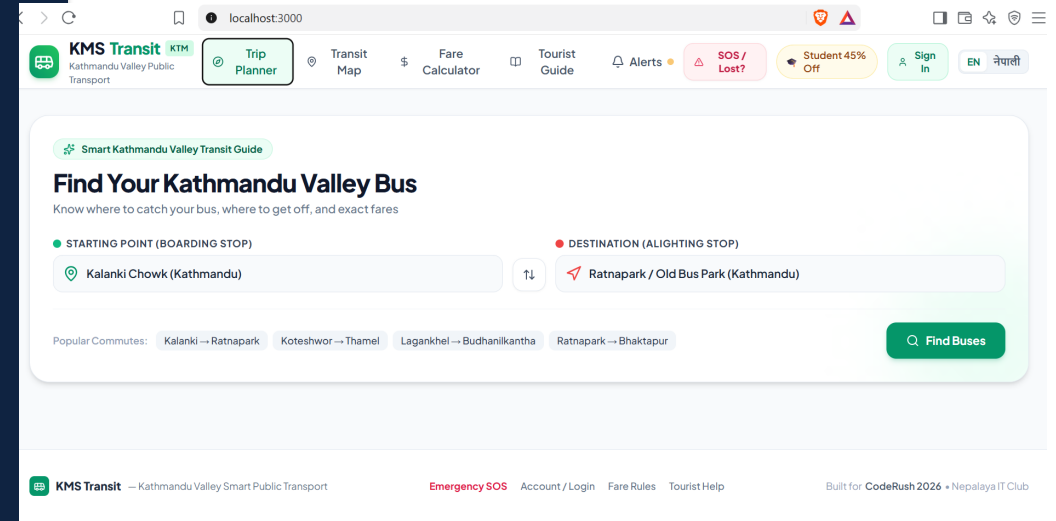


KMS Transit

Kathmandu Valley Smart Public Transport Guide

PROJECT PRESENTATION

Web-based public transportation assistant
CODE RUSH



React.js • Tailwind CSS • Modern Web UI

1. Introduction

A centralized digital guide for public transportation in Kathmandu Valley



- Kathmandu Valley has a large and complex public transportation network.
- Passengers can struggle to identify the correct bus, route, boarding stop and destination.
- Fare information and tourist-friendly transport guidance are not always easy to find.
- KMS Transit brings essential transportation services together in one simple web application.

2. Problem Statement

Key user problems addressed by the project



Finding the right bus

Users may not know which bus or route connects two locations.



Unclear stops

New passengers can be unsure where to board or get off.



Fare uncertainty

Passengers may need quick access to fare information.



Tourist difficulty

Visitors need simple, understandable local transport guidance.



Scattered information

Different transport needs are often handled separately.



Emergency needs

Lost-item and emergency information should be easy to access.

3. Objectives

What KMS Transit is designed to achieve

- Help users find suitable buses between two locations.
- Provide clear boarding and destination information.
- Make transit routes and stops easier to understand.
- Provide fare calculation and student discount information.
- Support tourists with practical transportation guidance.
- Provide quick access to alerts and SOS / lost assistance.
- Offer a simple bilingual interface in English and Nepali.

4. Technologies Used

Frontend technologies and development tools

React.js

Builds the interactive user interface using reusable components.

JavaScript

Handles application logic, events and user interactions.

Tailwind CSS

Provides responsive, utility-based styling.

HTML

Provides the structure of the web application.

Git & GitHub

Supports version control and project collaboration.

Vercel

Can be used to deploy the web application.

5. Main Features

The application combines multiple transport-related services



Trip Planner

Choose a starting point and destination to search for buses.



Transit Map

Explore stops and transportation routes.



Fare Calculator

Get transportation fare information.



Tourist Guide

Help visitors understand local transport.



Alerts

Access important transportation notifications.



SOS / Lost

Quick access to emergency and lost-item support.



Student Discount

Highlights the 45% student discount option.



Language

Switch between English and Nepali.

6. User Interface

Actual KMS Transit Trip Planner interface

The screenshot displays the KMS Transit Trip Planner interface. At the top is a navigation bar with the KMS Transit logo, a 'Trip Planner' button, and links for Transit Map, Fare Calculator, Tourist Guide, Alerts, SOS/Lost?, Student 45% Off, Sign In, and language options (EN, नेपाली). The main section is titled 'Find Your Kathmandu Valley Bus' with the subtitle 'Know where to catch your bus, where to get off, and exact fares'. It features two input fields: 'STARTING POINT (BOARDING STOP)' with 'Kalanki Chowk (Kathmandu)' and 'DESTINATION (ALIGHTING STOP)' with 'Ratnapark / Old Bus Park (Kathmandu)'. Below these are 'Popular Commutes' buttons for various routes. A large green 'Find Buses' button is at the bottom right.

KMS Transit KTM

Kathmandu Valley Public Transport

Trip Planner

Transit Map

Fare Calculator

Tourist Guide

Alerts

SOS / Lost?

Student 45% Off

Sign In

EN नेपाली

Smart Kathmandu Valley Transit Guide

Find Your Kathmandu Valley Bus

Know where to catch your bus, where to get off, and exact fares

● STARTING POINT (BOARDING STOP)

● DESTINATION (ALIGHTING STOP)

Kalanki Chowk (Kathmandu)

↑↓

Ratnapark / Old Bus Park (Kathmandu)

Popular Commutes:

Kalanki → Ratnapark

Koteshwor → Thamel

Lagankhel → Budhanilkantha

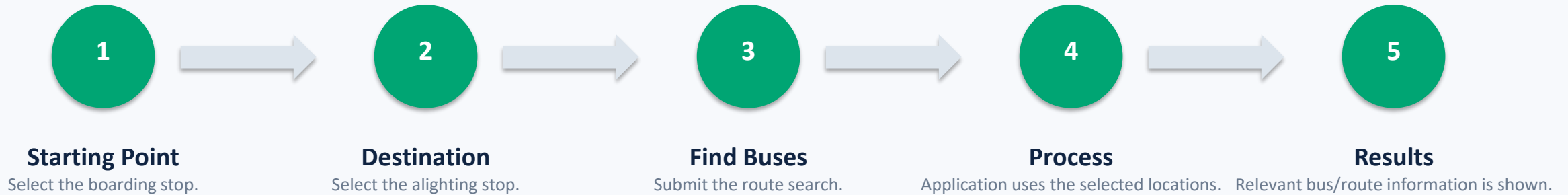
Ratnapark → Bhaktapur

Find Buses

The homepage emphasizes a simple flow: select boarding stop → select destination → Find Buses.

7. Trip Planner Workflow

A simple journey from user input to transport information



Example: Kalanki Chowk → Ratnapark / Old Bus Park

8. Navigation & User Experience

Designed to keep important functions visible and easy to reach



Trip Planner

Plan a journey



Transit Map

View routes/stops



Fare Calculator

Check fares



Tourist Guide

Help visitors



Alerts

See notifications



SOS / Lost

Emergency support



Student 45% Off

Discount information



EN / नेपाली

Language selection

9. UI/UX Design Principles

Focus on clarity, accessibility and quick decision-making

- Clean layout with clear visual hierarchy.
- Large, recognizable icons for transport functions.
- Prominent Find Buses call-to-action.
- Consistent green theme associated with transport and mobility.
- Clear distinction between boarding and destination fields.
- Popular commute shortcuts reduce repetitive input.
- Responsive design can support different screen sizes.

Design priorities

- ✓ Easy to understand
- ✓ Fast to use
- ✓ Visually consistent
- ✓ Accessible information

10. Testing & Validation

Checking the core user flows and interface

Test Case	Expected Result	Status
Application loads	Homepage appears correctly	PASS
Stop selection	Starting point can be selected	PASS
Destination selection	Destination can be selected	PASS
Find Buses	Search action is available	PASS
Navigation	Main service sections are accessible	PASS
Language	English / Nepali option is available	PASS
Responsive UI	Layout is designed for different screens	PASS

11. Future Enhancements & Conclusion

Possible next steps for turning KMS Transit into a richer smart-transit platform

- Real-time GPS bus tracking and live vehicle locations.
- Estimated bus arrival times and live service alerts.
- Interactive route navigation and optimized journey suggestions.
- Digital ticketing and online fare/payment support.
- AI travel assistant for natural-language route questions.
- Dedicated Android/iOS mobile application.

KMS Transit brings essential Kathmandu Valley public transport information into one user

Conclusion

Smarter • Simpler • More Accessible

Thank You

KMS Transit

Kathmandu Valley Smart Public Transport Guide